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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Retail Business Performance Dashboard Using Power BI

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Experiment No.: 3

Gith hub link: <https://github.com/kandlayasasri/Business-Analytics>

Retail Business Performance Dashboard Using Power BI

Problem Statement

A retail company relies on manually prepared reports to monitor business performance. This makes it difficult and time-consuming to analyze sales, revenue, profit, and product performance. Therefore, a Power BI dashboard is developed using the Superstore retail sales dataset to provide interactive visualizations and support data-driven decision-making.

Steps Followed

Step 1: Data Collection

- Obtained Sales Data from Excel files.
- Obtained Customer Data from CSV files.
- Obtained Product Data from My SQL Database.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded Excel, CSV files and My SQL data into Power BI.

Step 3: Data Cleaning and Transformation

- Removed duplicate records.
- Handled missing values.
- Renamed columns appropriately.
- Converted data types where necessary.

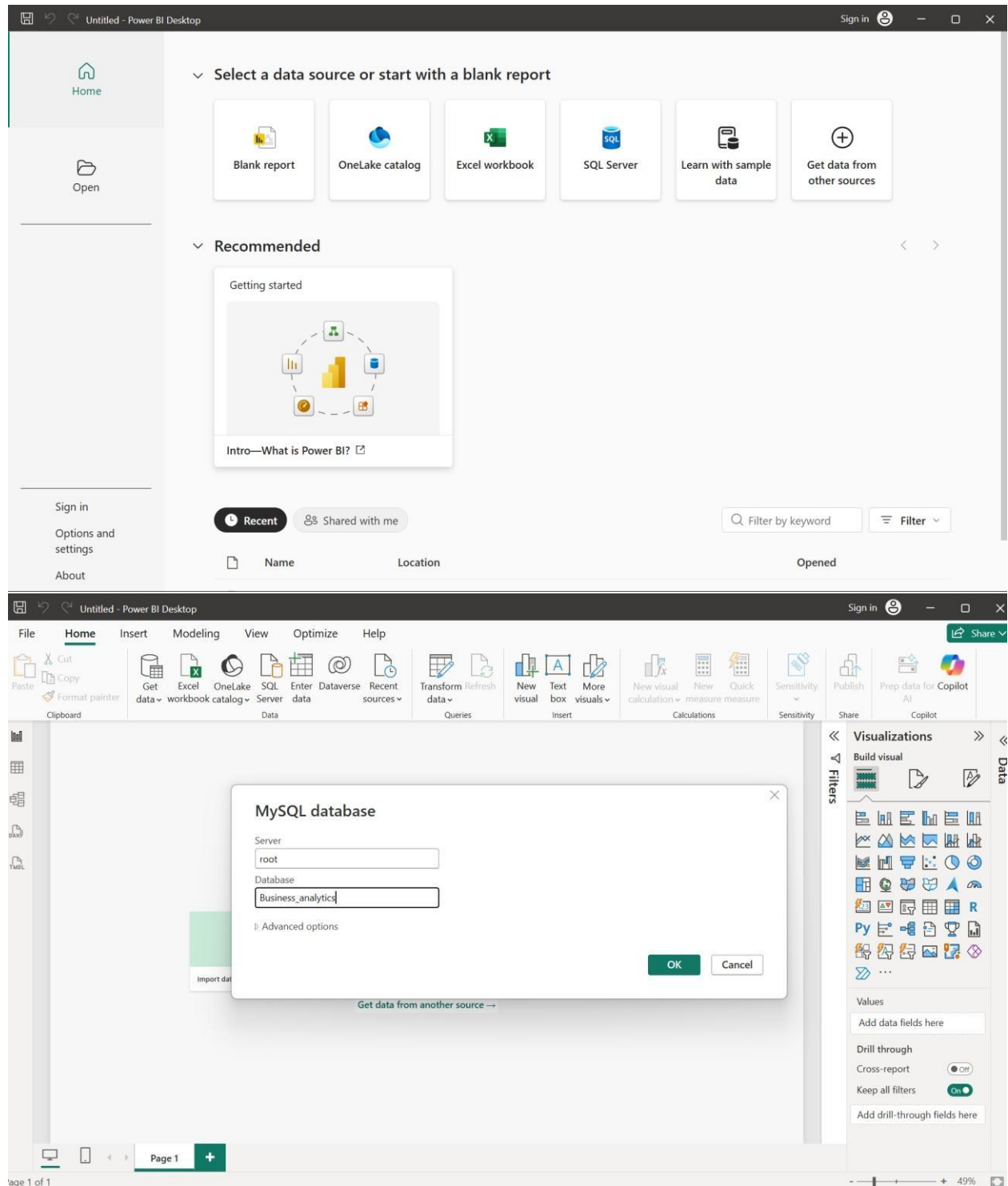
Step 4: Data Modeling

- Created relationships between Sales, Customer, and Product tables.
- Verified one-to-many relationships.

Step 5: Dashboard Development

- Created KPI cards for Total Sales, Orders, and Customers.
- Developed charts for Sales by Category and Region.
- Added slicers for Month, Category, and Region.
- Designed an interactive business dashboard.

Output



Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Combine

Queries [3] Sheet1 Customers_Data business_analytics produ...

= Table.TransformColumnTypes(#"Promoted Headers",{{"OrderID", type text},

	OrderID	OrderDate	CustomerID	ProductID
24	ORD024	46110	C004	P008
25	ORD025	46130	C005	P003
26	ORD026	46165	C006	P006
27	ORD027	46200	C007	P001
28	ORD028	46171	C008	P004
29	ORD029	46046	C009	P007
30	ORD030	46201	C010	P002
31	ORD031	46202	C001	P005
32	ORD032	46095	C002	P008
33	ORD033	46028	C003	P003
34	ORD034	46066	C004	P006
35	ORD035	46038	C005	P001
36	ORD036	46056	C006	P004
37	ORD037	46123	C007	P007
38	ORD038	46065	C008	P002
39	ORD039	46163	C009	P005
40	ORD040	46133	C010	P008
41	ORD041	46114	C001	P003
42	ORD042	46061	C002	P006
43	ORD043	46061	C003	P001
44	ORD044	46026	C004	P004
45	ORD045	46090	C005	P007
46	ORD046	46060	C006	P002

Query Settings

PROPERTIES

Name Sheet1

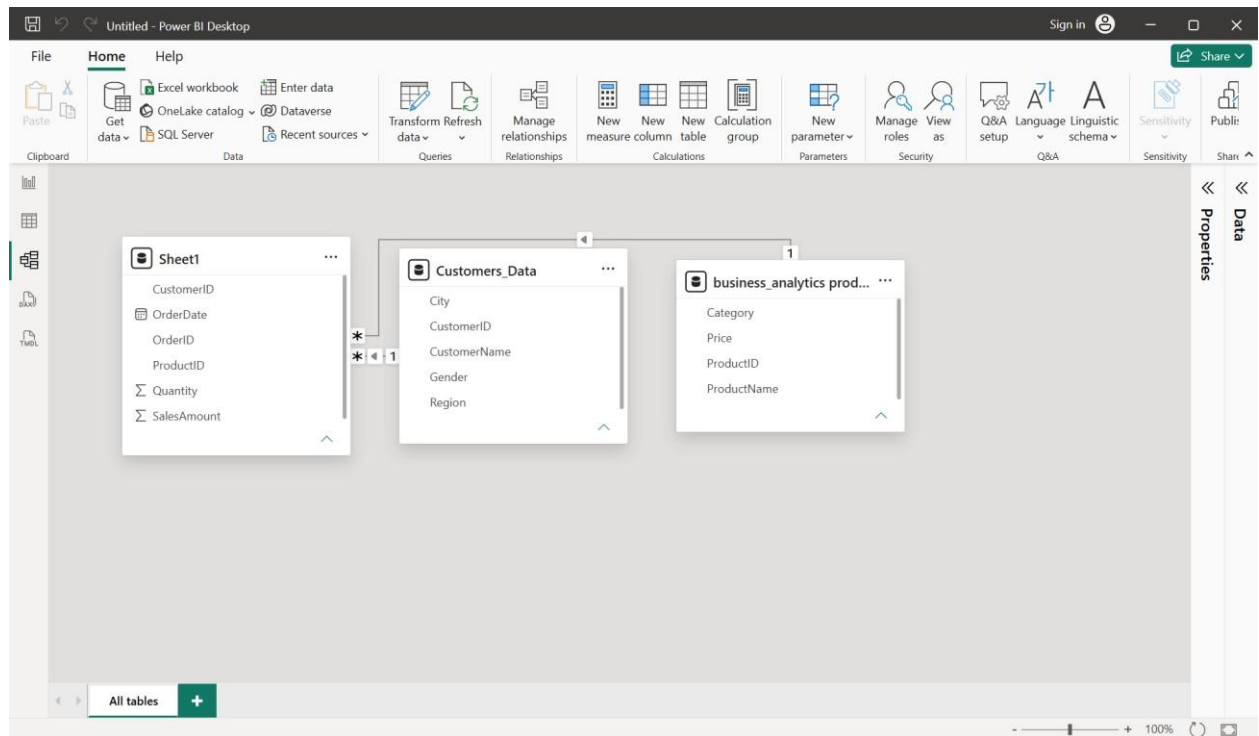
APPLIED STEPS

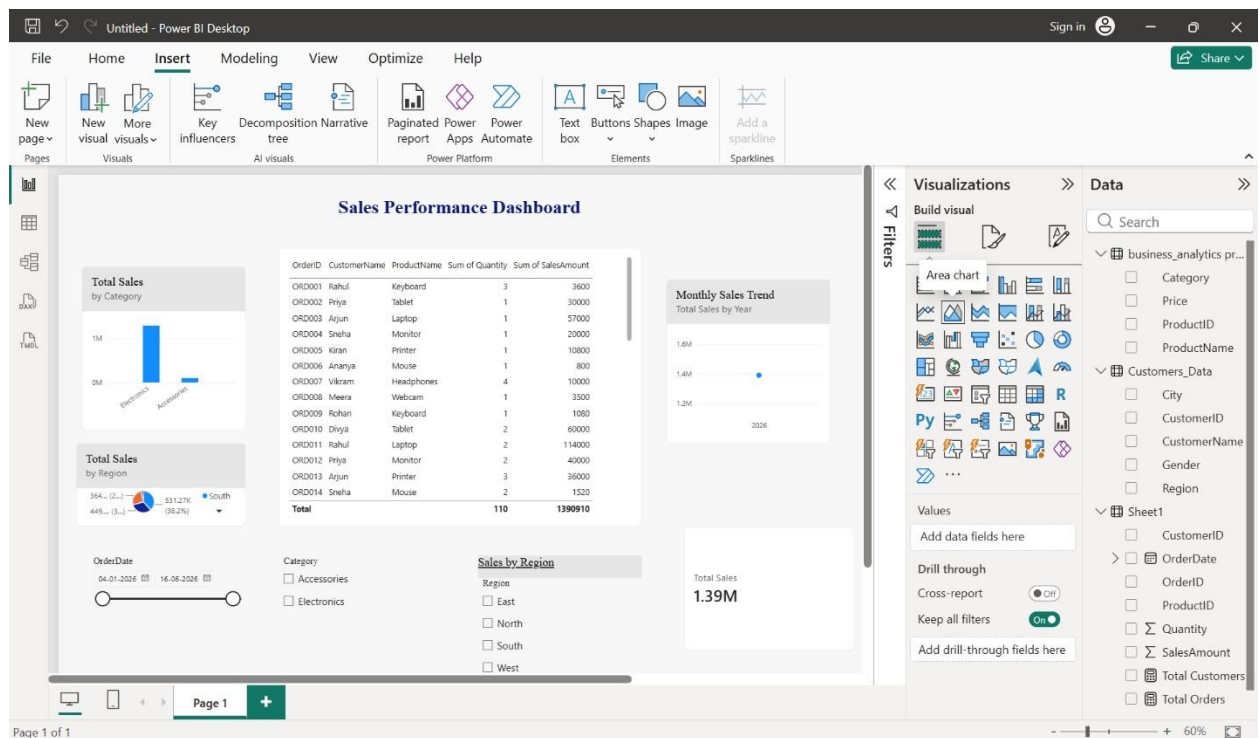
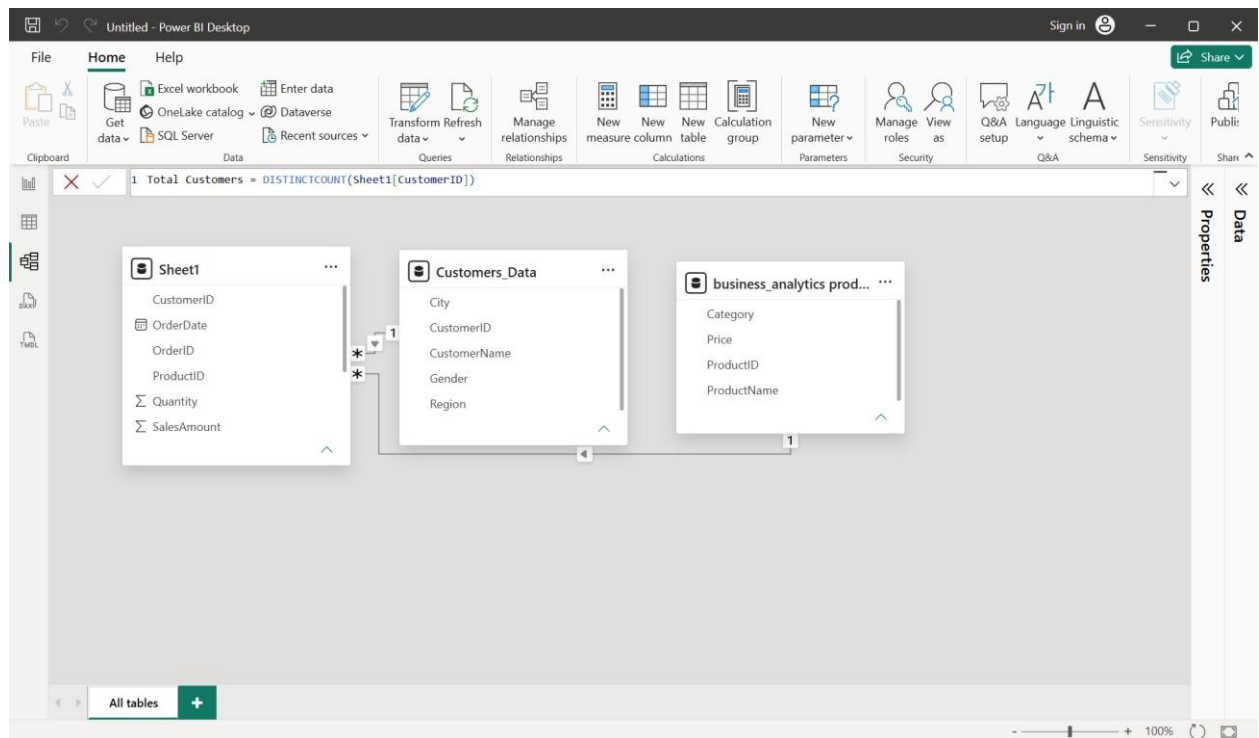
Source

Navigation

Promoted Headers

Changed Type





The final Power BI dashboard provides:

- Total Sales Performance
- Customer Analysis
- Product-wise Sales Analysis
- Regional Sales Performance
- Monthly Sales Trends
- Interactive Filters and Slicers

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Conclusion

This project successfully integrated data from multiple sources, including Excel, CSV, and MySQL, into Power BI and transformed the raw data into meaningful business insights. Data cleaning and transformation techniques were applied to improve data quality, including checking data types, handling missing values, and ensuring consistency across the datasets.

The Sales, Customers, and Products tables were connected using appropriate relationships, creating a structured data model for analysis. The developed Power BI dashboard provides a centralized view of business performance through key metrics and visualizations such as total sales, orders, customer information, category-wise performance, regional sales, and monthly sales trends.

Overall, this project provided practical knowledge of data integration, data preparation, data modeling, and business intelligence using Power BI. The resulting dashboard makes the data easier to understand and supports effective analysis and better data-driven decision-making.

Learning Outcomes

Through this project, I learned:

- Understand how to import and integrate data from multiple sources, including Excel, CSV, and MySQL, into Power BI.
- Learn how to clean and transform raw data by changing data types, handling missing values, and checking for duplicate records.
- Understand the importance of data modeling and create appropriate relationships between Sales, Customers, and Products tables.
- Gain practical knowledge of creating DAX measures such as Total Sales, Total Orders, and Total Customers.

- Learn how to create and use interactive visualizations, charts, KPIs, and slicers in Power BI.
- Develop the ability to analyze sales performance, customer information, product categories, regional performance, and monthly trends.
- Understand how business data can be transformed into meaningful insights to support data-driven decision-making.
- Gain practical experience in using Power BI as a business intelligence and data visualization tool.